

Chapter 2

Linear Algebra

Linear algebra is a branch of mathematics that is widely used throughout science and engineering. Yet because linear algebra is a form of continuous rather than discrete mathematics, many computer scientists have little experience with it. A good understanding of linear algebra is essential for understanding and working with many machine learning algorithms, especially deep learning algorithms. We therefore precede our introduction to deep learning with a focused presentation of the key linear algebra prerequisites.

If you are already familiar with linear algebra, feel free to skip this chapter. If you have previous experience with these concepts but need a detailed reference sheet to review key formulas, we recommend *The Matrix Cookbook* ([Petersen and Pedersen, 2006](#)). If you have had no exposure at all to linear algebra, this chapter will teach you enough to read this book, but we highly recommend that you also consult another resource focused exclusively on teaching linear algebra, such as [Shilov \(1977\)](#). This chapter completely omits many important linear algebra topics that are not essential for understanding deep learning.

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